

$\Rightarrow * \sim \text{Scalar (1-D)}$

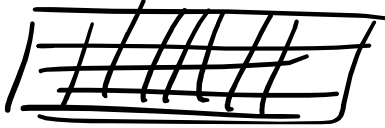
$\Rightarrow \text{---} \sim \text{Vector (2-D)}$

$\Rightarrow \begin{array}{c} \uparrow \\ \swarrow \quad \searrow \end{array} \sim \text{Matrix (3-D)}$

$\Rightarrow \begin{array}{c} \text{Time} \\ \curvearrowright \end{array} \begin{array}{c} \uparrow \\ \swarrow \quad \searrow \end{array} \sim \text{Tensor (N-D)}$

$\Rightarrow$ With the perspective of CS

$\hookrightarrow$ Vector is single column. 

$\hookrightarrow$ 2 columns are a matrix. 

$\hookrightarrow$ N-columns are tensor.



$\Rightarrow X \sim \text{independent variable is a matrix}$

$\Rightarrow X \sim$ independent variable is a matrix

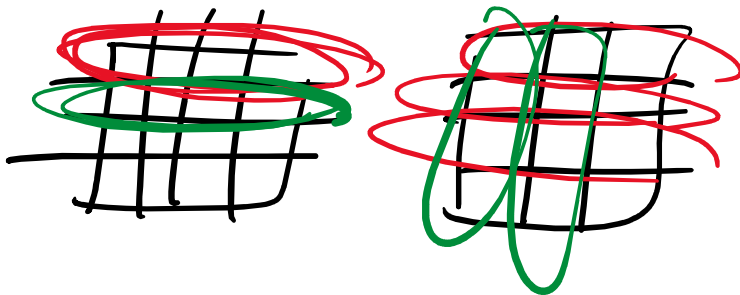
$y \sim$ dependent variable is always a vector.

$\Rightarrow$ Method: `np.shape [100, 1]`

$$\sum_{i=1}^N \frac{(x_i - \bar{x}) \cdot (y_i - \bar{y})}{(x_i - \bar{x})^2} \quad \begin{array}{c} O(n) \\ + \\ O(1) \\ \hline O(n) \end{array}$$

Recurring (Loop)

$\Rightarrow$ Vectorization: (Dot product) $(* / \cdot)$

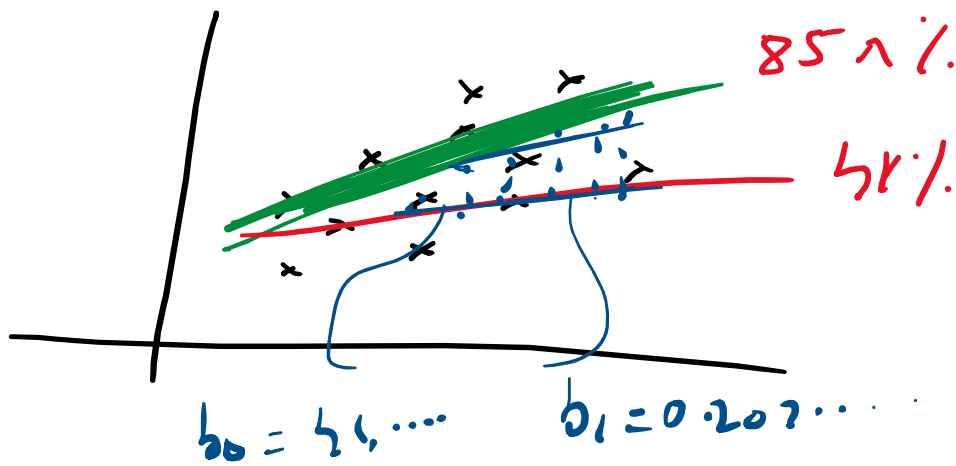


$\Rightarrow$ Flatten:

$$(n, 2) \quad \begin{bmatrix} [] \\ [] \\ [] \end{bmatrix} \quad \left| \begin{bmatrix} [] \\ [] \\ [] \end{bmatrix} \right. = \quad \begin{bmatrix} - \\ - \end{bmatrix} \quad \left| \quad (n, 1) \right.$$

$$(n, 2) \quad \begin{bmatrix} x_1 & x_2 \\ x_1 & x_2 \\ x_1 & x_2 \\ x_1 & x_2 \end{bmatrix} \quad \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} - \\ - \\ - \\ - \end{bmatrix} \quad \begin{bmatrix} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \\ \begin{bmatrix} x_1 & x_2 \end{bmatrix} \\ \begin{bmatrix} x_1 & x_2 \end{bmatrix} \\ \begin{bmatrix} x_1 & x_2 \end{bmatrix} \end{bmatrix}$$

$\Rightarrow$ Optimization : Re-training the parameters of the model.



R^2
Slope β
Intercept are parameters

$\Rightarrow$ Gradient Descent